

TableCraft

Automated Multi Category Data Formatting Web Application

Project Book and Step-by-Step Guide

Prepared for: BCA Semester 6 College Project Submission

College: GK CK Bosamia College, Jetpur

Project Name: TableCraft - an Automated Multi Category Data Formatting Web Application

Main Concept: User types data, the website formats it into a table and supports download in PDF, DOC and PNG formats.

Tech Stack: HTML5, CSS, Bootstrap, JavaScript, PHP and MySQL on WAMP Server.

This document explains the project idea, features, database design, installation flow and live demo workflow for the college presentation.

1. Introduction

- TableCraft is a responsive web application designed to help users convert raw typed data into clean table layouts.
- The project is suitable for academic, commercial and office records such as student data, employee records, shop orders and library registers.
- The main goal is to reduce manual formatting effort and provide quick export options.

2. Objectives

- Allow users to register and log in securely.
- Generate structured tables from simple typed input.
- Support popular categories and custom categories.
- Provide edit, update and delete workflows for table records.
- Enable professional download formats for presentation and reporting.

3. Core Modules

- Authentication module for register and login.
- Dashboard module with animated statistics and category cards.
- Table builder module for category-based input and preview.
- Custom category editor for user-defined column sets.
- Export module for PDF, PNG and DOC downloads.
- Theme switch module for bright and dark display modes.

4. Popular Categories

- Student Record
- Shop Order
- Employee Info
- Library Book List
- Product Stock
- Commercial Category
- Custom Category Editor

5. Folder Structure

The project is organized in a simple and readable structure for WAMP hosting.

Folder / File	Purpose
auth/	Login, register and logout pages
assets/css/	Responsive styling and theme colors
assets/js/	Theme toggle, canvas animation and export logic
config/db.php	Database connection file
includes/	Shared bootstrap, header, footer and helper functions
tables/	Create, view and delete table pages
custom/	Custom category editor
database/tablecraft.sql	Database schema and sample setup
TableCraft_Project_Guide.pdf	Project book guide

6. Database Design

The application uses MySQL tables for authentication, custom categories and saved table records.

- **users** - stores name, email and hashed password.
- **custom_categories** - stores reusable custom field sets for the logged-in user.
- **saved_tables** - stores table title, category, columns, row data, notes and timestamps.

The database script is included in **database/tablecraft.sql**. Import it into phpMyAdmin before running the app.

A demo user is included for quick testing. You can also create a new account using the registration page.

7. Installation Steps on WAMP

- Copy the TableCraft_Project folder into the WAMP www directory.
- Start Apache and MySQL from the WAMP tray icon.
- Open phpMyAdmin and import database/tablecraft.sql.
- Verify database credentials in config/db.php.
- Open the project in browser using localhost/TableCraft_Project/.

Suggested demo flow for viva or presentation:

- Show login and register screens.
- Open the dashboard and explain the category cards.
- Create a table using one category.
- Edit a row and delete a row in preview.
- Save the table and open the export buttons.
- Demonstrate PDF, PNG and DOC downloads.
- Create one custom category and reuse it.

8. User Flow

- Register or login.
- Choose category from dashboard or builder.
- Paste data using pipe, comma or tab separators.
- Preview, edit and save the table.
- Open saved table and export it in different formats.

9. Security and Validation

- Password hashing is used during registration.
- SQL queries are written with prepared statements.
- Session-based login protects private pages.
- Input fields are validated on the server side.
- Theme selection is stored in a browser cookie.

10. Future Scope

- Add more category templates.
- Add drag and drop table builder.
- Add server-side PDF generation library.
- Add multi-user sharing and role-based access.
- Add charts, analytics and advanced filters.

11. Conclusion

TableCraft is a practical college project that combines table generation, customization, authentication and export functionality in one clean web application. The project is suitable for demonstration as well as documentation in the final project book.

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